

Week 5

What you will have to achieve this week

This week is all about **community finding algorithms**, in particular about the Louvain algorithm.

You will write a programme which partitions a network into separate communities, identifies a community which contains a target protein, and identifies communities adjacent to it.

You will use the Louvain algorithm implemented in network as the subroutine `louvain_communities`. This routine aims to find an approximation of a partition that maximizes *modularity*. Modularity is essentially a measure of how densely connected some groups of nodes are to each other relative to the rest of the network (not quite like that, but we will discuss this in detail in the lectures a bit later).

The routine contains a resolution parameter you can use to have some control over the sizes of the communities, i.e. the number of nodes contained in the communities. How to use the resolution parameter is described in the documentation.

Write subroutines which

- partition a given connected graph into communities
- sort the communities after their size and show the number of communities and their size distribution
- eliminate all communities which are below a certain specified size (recall that we do not want to consider communities which contain only a single node); record again the number of communities and their size distribution
- identifies and stores those communities which are adjacent to a community containing a specified protein (which eventually will be the target protein given to you by your BCMB group). These adjacent communities you can then further analyse with the centrality measures developed last week

The Louvain algorithm (like most community finding algorithms) involves a random seed and hence its outputs will be different each time you run it. All these outputs will have similar values of their modularity but may differ in the allocation of some nodes to a community. For networks with a very clear community structure this is not an issue, however, for the large PPI network it is. Quantify the randomness by monitoring the number and sizes of the communities.

This is the Pass/Credit level. If you aim for distinction or higher distinction you can work through the sheets for the upcoming weeks.